Homework 7

Module 7b - PK-PD Relationship · Homework · No date

Name _____ Date _____

1. A 75-kg volunteer received a single IV bolus dose of an investigational drug. The drug was administered at a dose of 15 mg/kg. The drug has an apparent volume of distribution that is 0.3 L/kg and an elimination half-life approximately 4 hours .

1a What is the AUC following the IV bolus dose?

1b If the MEC is 3.75 mg/L, what is the duration of activity of this drug in this volunteer?

1c What is the duration of activity if the dose were 7.5 mg/kg?

1d What dose would provide a duration of action of 8 hours?

2. A 68-kg, 28-year-old female is receiving phenytoin for the treatment of seizures. When this patient was taking a daily oral dose of 300 mg, her steady state plasma concentration was 9.12 mg/L. The daily dose was increased to 400 mg which resulted

in a steady-state plasma concentration of 21.8 mg/L. Assume 100% bioavailability and an apparent volume of distribution that is equal to 70% of body weight.

2a What is this patient's KM?

2b What is the Vmax in this patient?

2c What is the daily dose rate of phenytoin that should achieve a steady state phenytoin plasma concentration of 15 mg/L in this patient? (Round to the nearest ten)

2d What is the half-life and clearance in this patient while she is taking the dose that you recommended above to achieve a steady state concentration of 15 mg/L? What is the half-life?

2d What is the half-life and clearance in this patient while she is taking the dose that you recommended above to achieve a steady state concentration of 15 mg/L? What is the clearance?

2e What is the expected steady state concentration if this patient were to take 325 mg/day?